How AI Revitalized the Spirit of Synthetic Geometry

By James Squires

I'm sure everyone occasionally has flashbacks to high school geometry–I do myself! I really didn't like school at the time and I mostly zoned out during class. However, when I actually did the math problems, there was a certain satisfaction that I received. When we were given in-class problems, I had to compensate for my lack of scholarly attitude by maximizing the utility I gained from the knowledge I did retain; I would typically try deriving any angle identity we were given from my understanding of right angles and parallel lines. In hindsight, I can say that my experience of having been a bad student was good for my education–this was likely my first exposure to the practice that mathematicians do; not computation, but compressing your knowledge into a few facts that can derive many things. To quote the philosopher Helvétius, *"The knowledge of certain principles easily compensates the lack of knowledge of certain facts."* This is the practice, virtue, and strength of mathematics.

The Birth of Mathematics

Geometry was one of the first mathematics developed. The etymology means "measurement of earth". Unsurprisingly, the first applications were largely for measuring land plots for such things as ownership. But around 300 B.C.E., something truly incredible happened. Euclid wrote his book *Elements*. Euclid, a Greek mathematician, developed what we call an *axiomatic* formulation for mathematics. He compiled many previously known results, and organized them into a cogent masterpiece where from five seemingly obvious statements, we can derive many of the iconic geometry theorems we know and love. Doing so streamlines mathematical progression by establishing not only a common language, but a common set of facts with which we can build on. Furthermore, it mitigates the possibility of contradiction while increasing the possibility of novel discovery. When we know what the limits of our system of mathematical thought are, we know where the frontier lies which then tells us what we don't know. And when we force our mathematics to be organized systematically, it's easier to pinpoint where a deviation or misstep happens. Thus, Euclid didn't *just* axiomatize mathematics, I'd argue he developed the canonical mathematical text that would go on to inspire essentially all of modern mathematics in one way or another.

From classical Greece through the middle ages, a standardized education consisted of four areas of study: geometry, arithmetic, music, and astronomy. Even before Euclid, geometry was viewed as a fundamental pillar of knowledge. And throughout the Renaissance up to the late 1800s, mathematicians were expected to learn Greek explicitly in order to read Euclid's Elements, the magnum opus of early mathematics. The book has the spirit of the field itself. However, because the book was so well studied (it's one of the most published books ever [1]) people noticed a couple of logical 'holes'. To fix this, a mathematical physicist named David Hilbert developed a modern axiomatic formulation with an enhanced level of rigor. The style of axiomatization he used has inspired what we now call *Hilbert Systems* in mathematics, systems of logic largely based on a choice of axioms.

At around the same time, other mathematicians developed our foundational logic called set theory. And then a couple decades later more revolutionary bounds were made in *Model Theory*, the study of the entire mathematical systems formed from the axioms and the rules that we choose. But a subtle shift began happening that only became obvious in the past few decades. As we made these greater advances, spurred by Euclid's attempt to systematize geometric understanding, the use of geometric intuition declined. Specifically the use of diagrammatic reasoning, like when you actually draw geometry to form a mathematical proof, declined. Instead, we began to rely more and more on analytic methods of study. By analytic, I mean math with coordinates that can be computed. This slowly shifted the focus of mathematics from what is called *synthetic geometry*, geometry of objects that are their own thing (not defined with numbers), to *analytic geometry* which defines objects through coordinate systems and analytical properties. An example from synthetic geometry is how we say a line contains at least three points, whereas in analytic geometry we might say the line y = 0 contains (0, 0), (1, 0), and (-1, 0).

Analytic and Synthetic Geometry

Analytic geometry has its obvious merits. In the real world, we measure things! You can't do physics without measuring different physical quantities such as length, mass, time, or velocity. When we measure things, we obtain values that we must analyze numerically. When Newton developed calculus to describe classical mechanics–how forces in the macroscopic world interact–he developed an analytic discipline. When we study the geometry of objects, we use coordinate systems because they allow us to calculate properties and effectively manipulate these objects. Quite often, it's far easier to perform a task when using analysis. That being said, we can still opine the decline of synthetic geometry, the beautiful pearl of mathematics. One of the first fields, the field that coupled mathematical thought to intuition, the field that imbued logic and deduction into mathematics, is now less and less valued.

Furthermore, the decline of synthetic geometry isn't without its consequences. Diagrammatic reasoning with ruler-and-compass is an amazing heuristic for solving mathematical problems. It's not difficult to claim that, to an extent, the geometric intuition previously held by almost all mathematicians has atrophied into a select skill. But something weird is happening.

Automatic Theorem Proving, AI, and Geometry

This article is not about AI, but rather about a consequence of AI. It is well known that Large Language Models (LLMs) can hallucinate and quite often make mistakes. But it is also well known that they can frequently surprise us with quite insightful ideas into particular domains of study. So it seems we have a machine with potential for positive use, but also the risk of hard-to-detect malfunctions. Interestingly enough, we can actually mitigate this problem through the use of a recent advancement in logic.

Coding languages for mathematics have developed in the past two decades into a booming field of study. These languages allow us to do mathematical proofs in a computer in such a way that if an error is made, the computer detects said error. This means if you define a theorem correctly, then you can check someone's mathematical proof of the theorem immediately (if the proof is put in terms of this code). So what happens if we do this with the mathematical proofs we ask LLMs to produce? We can have proofs of theorems generated almost instantly and checked for correctness. We throw out the ones that fail and keep the rest. This gives us a fully automated way to solve for the proofs of mathematical theorems. Although AI can't always solve mathematical theorems, as there are limitations on the thinking capacity of AI (or lack thereof), this still gives a means of *Automatic Proof Search*. But to further develop this area of research, we may want to try this method out and refine it on particular areas of mathematics. Well, what mathematical branch of study has been thoroughly explored, is logically organized in a way that we can code it into a computer, is sufficiently complex and interesting, and basically forms a playbox for AI to explore mathematical proofs in? Synthetic geometry!

Recent consolidated efforts have seen the development of benchmarks and models for Automated Theorem Proving of geometry, and they are growing better each year. It's not an overstatement to say this is likely an underground revolution for mathematics. This will change the paradigm for how mathematicians do math (I intend to discuss this in a future article, I don't think this is a bad thing). But I find something highly poetic about this development: the mathematical formalism that catalyzed the modern field as we know it has staged a return as one of the first mathematics to be mathematically formalized through AI.

References:

Autoformalizing Euclidean Geometry: Autoformalizing Euclidean Geometry

LeanDojo (automatic theorem proving with LLMs): 2306.15626